1. Purpose

1.1. The purpose of this documentation is to define test method of tire electric resistance.

1.2. This is an extension of the engineering drawing. All parts should be satisfied all requirements, parts size and relevant requirements at one time.

2. Application of scope

2.1. These regulations describe the test methods and requirements of tire electric resistance for vehicle.

2.2. If the drawing instruction is different from this Spec. drawing instruction is prior.

2.3. If there is any discrepancy between this standard of Korean version and English version, Korean version will take precedence.

2.4. All specimens, which are randomly selected, examined parts from the proto process and the production process (including pilot process), must satisfy this standard to carry out the specified process in this document. Relevant supplier should establish a reliability verification system themselves to qualify the design of the part with sufficient number of samples.

3. Terminology

3.1. Tire electrical resistance (in ohms) means the electrical resistance measured between the wheel of a mounted and inflated tire-wheel assembly and a metallic plate onto which the tire is loaded at a specific load.

4. Mark on the drawing

The drawing indication of this standard follows as below.
[Tire electrical resistance must conform to ES 52931-03]

5. Requirement

The measured electrical resistance should be lower or equal to 10^8 Ω in order to dissipate the vehicle static charge easily.

※ NOTE : LEGAL RESPONSIBILITY

[This specification defines only the least requirement of tire.
Radically Tire Suppliers guarantee the tire robustness for vehicle static charge.]

Test item		Criterion	Test method	Remark
Electrical resistance test	Electrical resistance	Among the resistance values measured at least three circumferential locations evenly spaced around the tire, the maximum value should be lower or equal $10^8 \Omega$. ※ Exceptionally, The maximum electrical resistance of tire for Fuel-cell Electrical Vehicle which using Hydrogen for fuel should be lower or equal $2.5 \times 10^7 \Omega$.	6	TABLE 1 FIG.1

6. TEST PROCEDURE

6.1. Apparatus

6.1.1. Resistance Measuring Instrument (ohmmeter)

Resistance shall be measured by a commercial instrument capable of measuring electrical resistance in ohms and having a power source capable of 1000 V. The voltage shall be controlled as described in Table 1 and shall not dissipate more than 3 W in the test sample. The instrument shall be capable of determining the resistance up to a value of 10^{12} Ω with an accuracy of ±5 %. The input impedance shall be at least 10^{14} Ω.

TABLE 1 Test Voltage

Tire Resistance Range (Ω)	Test Voltage (V)
10^3 to 10^4	1
10^4 to 10^6	10
10^6 to 10^8	100
10^8 to 10^{12}	1000

6.1.2. Metal Loading Plate

A flat plate of dimensions sufficient to encompass the entire contact surface of the tire under test and with sufficient thickness to support the test loads without visible deformation. This plate shall be made of a conductive noncorrosive metal, for example, brass or stainless steel, free from any coating or obvious surface contamination, such as oxidation or corrosion. Aluminum shall not be used for the plate because of its high susceptibility to the rapid development of surface oxides, which may adversely affect reading accuracy.

6.1.3. Loading Apparatus

A loading fixture (Fig. 1) capable of applying the tire load, in a radial direction, against the metal loading plate. Test load measurement accuracy shall be ±1%.

6.1.4. Insulating Material

A sheet of insulating material such as polyethylene, PTFE (polytetrafluoroethylene), or equivalent, with sufficient strength to support the test loads without visible deformation. The insulating material should have dimensions of at least 50 mm greater, on all sides, than the metal loading plate.

6.1.5. Pressure Gage

A commercially available gage with an accuracy of ±3 kPa.

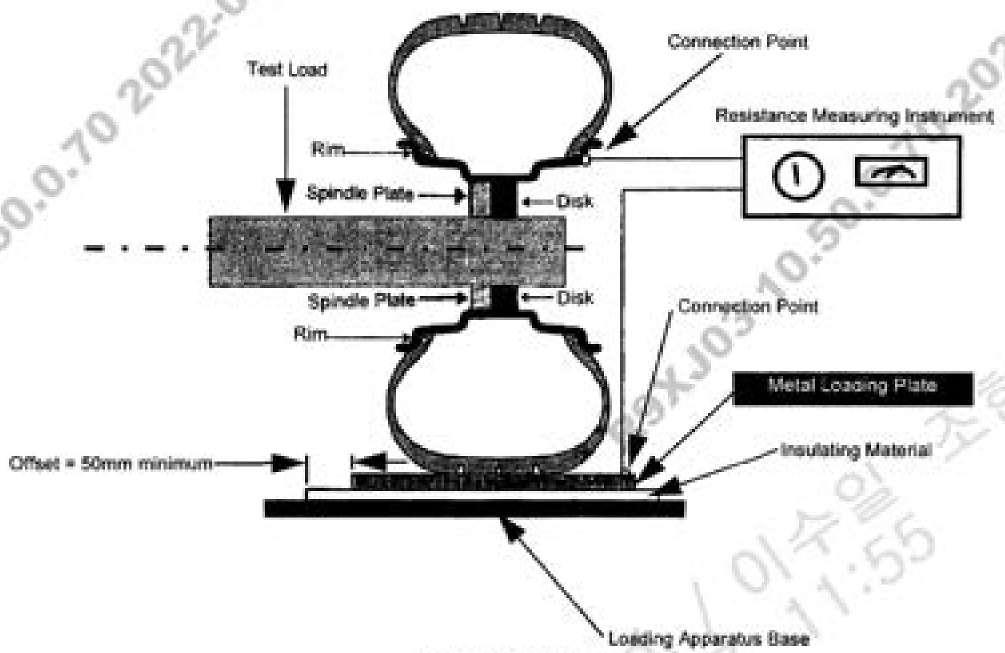

FIG. 1 Apparatus

6.2. Conditioning

For at least 8 h prior to measurement of passenger car tire, the tire to be tested shall be kept at an ambient temperature of 23 ± 5°C [73 ± 9°F], and at a relative humidity less than 60 %.

6.3. Measurement Conditions

6.3.1. The test load applied during the measurement is 80 ± 5 % of the maximum load capacity of the tire as listed in the applicable TRA, ETRTO or JATMA yearbook.

6.3.2. The inflation pressure is equal to 80 ± 5 % of the pressure corresponding to the maximum load of the tire.

6.3.3. If the tire size is not listed in the applicable TRA, ETRTO, or JATMA yearbook, the above percentages apply to the loads and inflations as marked on the sidewall of the tire.

6.3.4. Ambient temperature during the measurement shall be maintained at 23 ± 5°C (73 ± 9°F).

6.3.5. Relative humidity during the measurement shall be maintained at ≤ 60 %.

6.4. Procedure

6.4.1. Preparation of the Tire-Wheel Assembly

- The approved wheel (steel preferred) must be stripped clean in the bead seat area, as well as at the connection point. As an alternative, if The electrical resistance of The wheel is known to be two orders of magnitude lower than The tire to be measured, stripping is not necessary.
- It is necessary to make sure that the tire is dry before taking the measurement. dry mount the tire if possible. to avoid damage to tires in the case of difficult mounting conditions, a water-soluble mounting solution can be used. Any mounting solution on the sidewall or tread of The tire must be cleaned and dried.
- Mark a reference point on The tire sidewall with a nonconductive material. if the tire has a DOT marking, this should be used as the reference point.

6.4.2. Setup

- Set up the apparatus as shown in Fig. 1.
- Clean the metal loading plate with isopropyl alcohol or a similar agent and allow to dry.
- Install The tire-wheel assembly on the loading fixture and clean the exterior of the tire with isopropyl alcohol or a similar agent and allow to dry. Conductive or nonconductive substances on the tire such as mold release agents, or paints, or both, which could affect the results, must be removed. The use of organic solvents likely to attack the rubber is prohibited.
- Connect the ohmmeter leads to the metal loading plate and to the wheel. Alternately, a connection can be made on the loading fixture if the resistance between this point and the bead seat of the wheel is no more than 10 Ω.

6.4.3. Preload Cycle

- Load the tire-wheel assembly at the reference point to the value specified in 6.3.1, hold for 1 min, and then remove the load.
- Repeat The load-unload cycle a second time.

6.4.4. Measurement

- If the range of the resistance is unknown, initially apply 1000 V and decrease the voltage according to Table 1 as necessary. Load the tire a third time to the test load and immediately apply the test voltage in accordance with Table 1.
- Record the resistance measurement 3 min ± 10 s after the voltage has been applied. The voltage and load are to be applied continuously to the tire until after the final measurement is recorded.
- Unload The tire.
- Repeat the preload cycle and measurement in accordance with 6.4.3. and 6.4.4. for at

least two additional circumferential locations evenly spaced around the tire.

6.5. Resultant Electrical Resistance of Tire

The resultant electrical resistance of the tire is the maximum of the electrical resistance measurements of all circumferential locations tested.

6.6. CHECK AND QUALITY CONTROL

- All the test item of this specification and required test item by HMC/KMC Engineering design team are satisfied per each test period is arranged under agreement with HMC/KMC quality checking team.

7. Relevant specifications

7.1. ASTM F1971-99

7.2. ETRTO

8. Attachment

8.1. Engineering Specification Test & Development Procedure (Attachment 1)

Attachment 1 개체-1

8.2. Engineering Specification Test Proposal (Attachment 2)

Attachment 2 개체-2

8.3. Engineering Specification Test & Development Report (Attachment 3)

Attachment 3 개체-3

Supplement

- This Annexed Document provides for required ES minimum valuation quantity

Annexed Document Table 1. Minimum Valuation Quantity

Evaluation Method No	Evaluation Item	Valuation Quantity (Amt of sample to be tested / Amt of sample to be recorded)
6	Electrical resistance test	2/2
		2/2
		2/2
		2/2
		2/2
		2/2
		2/2

* Supplier manages Mill sheet about constitutional parts and provides it to H/KMC
* Valuation item and sample quantity can be changed by requiring from H/KMC
  * Subcontractor has liability to adjust minimum quantity of valuation sample which is required for guaranteeing durability on annexed document table 1.